

Task 1:

DATA

```
students = [ ["Name", "Marks"], ["Ali", 85], ["Sara", 42], ["Ahmed", 45], ["Zain", 40], ["Ayesha", 61], ["Ayaan", 80], ["Danial", 71], ["Aliza", 50], ["Danish", 91], ["Hamza", 68], ["Fatima", 77], ["Usman", 59], ["Bilal", 88], ["Hina", 73], ["Noor", 95], ["Maham", 66], ["Saad", 54], ["Iqra", 82], ["Laiba", 47], ["Kashif", 79], ["Amna", 62], ["Umer", 35], ["Yasir", 84], ["Anaya", 58], ["Rehan", 93], ["Sana", 69], ["Farhan", 75], ["Areeba", 81], ["Hasan", 49], ["Muneeb", 87] ]
```

Using a **for loop** and **if-else statements**, display each student's:

- Name
- Marks
- Grade according to:
 - A: 80+
 - B: 70–79
 - C: 60–69
 - F: Below 60

Finally, print how many students passed and how many failed using frequency table.

Task 2:

Using Pandas. Use **data_grand_test.csv**

- Display first 5 rows
- Display last 5 rows
- Display data types
- Display total rows & columns
- Display frequency of Departments
- Display frequency of Status

Clean the following columns:

Name

- Remove leading/trailing spaces

City

- Remove spaces
- Convert to Upper Case (Like → KARACHI)

Status

- Using your knowledge of cleaning, clean the Status column. Show each step.

Skills

The Skills column contains values like

Python, Excel

Python ,SQL

Excel,Power BI

Tasks

- Remove unnecessary spaces using
- Split the column in Skill1,Skill2,...

Finally print the cleaned dataframe.

Task 3:

Some values in **Performance** are strings.

- Convert **Performance** into integer.
- Display employees having **Performance** greater than 80.
- Display employees from Karachi.
- Display employees from IT department having **Performance** less than 75.
- Display average salary.

Task 4:

Part A – Merge

1. Merge **employees.csv** and **salary.csv** using EmployeeID.
2. Merge the resulting dataframe with **performance.csv**.
3. Display the final dataframe containing:
 - EmployeeID
 - Name
 - Department
 - Salary
 - Rating

Part B – Concat

4. Concatenate **employees.csv** and **new_employees.csv** to create a single employee database.
5. Display the total number of employees after concatenation.

Part C – Analysis

Using the merged dataframe,

- Display employees whose salary is greater than **80,000**.
- Display employees with **Rating A**.
- Calculate the average salary department-wise.